

Data Report on Yield Curves

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This report analyzes/compares the U.S interest rate term structure between late May-June 2004 and current day. The two time periods reflect vastly different economic environments: the 2004 period shows a period of early-stage economic recovery from the dot-com bubble, while the 2025 period reflects a restrictive policy environment, characterized by an inverted yield curve and an uncertain financial climate in the future.

```
my_birthday<- "2004-05-30"
tickers<- c("DGS3M0","DGS2","DGS5","DGS10","DGS30")

yield_data_xts <- do.call(merge, lapply(tickers, function(x) {
  getSymbols(x, src = "FRED", auto.assign = FALSE)
}))

yield_df <- data.frame(Date = index(yield_data_xts), coredata(yield_data_xts))

colnames(yield_df) <- c("Date", "3M", "2Y", "5Y", "10Y", "30Y")

birth_row <- yield_df %>%
  filter(Date >= as.Date(my_birthday)) %>%
  filter(!is.na(`3M`)) %>%
  slice(1) %>%
  mutate(Label = "Birth Date")

current_row <- yield_df %>%
  drop_na(`3M`) %>%
  tail(1) %>%
  mutate(Label = "Current")

plot_data <- bind_rows(birth_row, current_row)

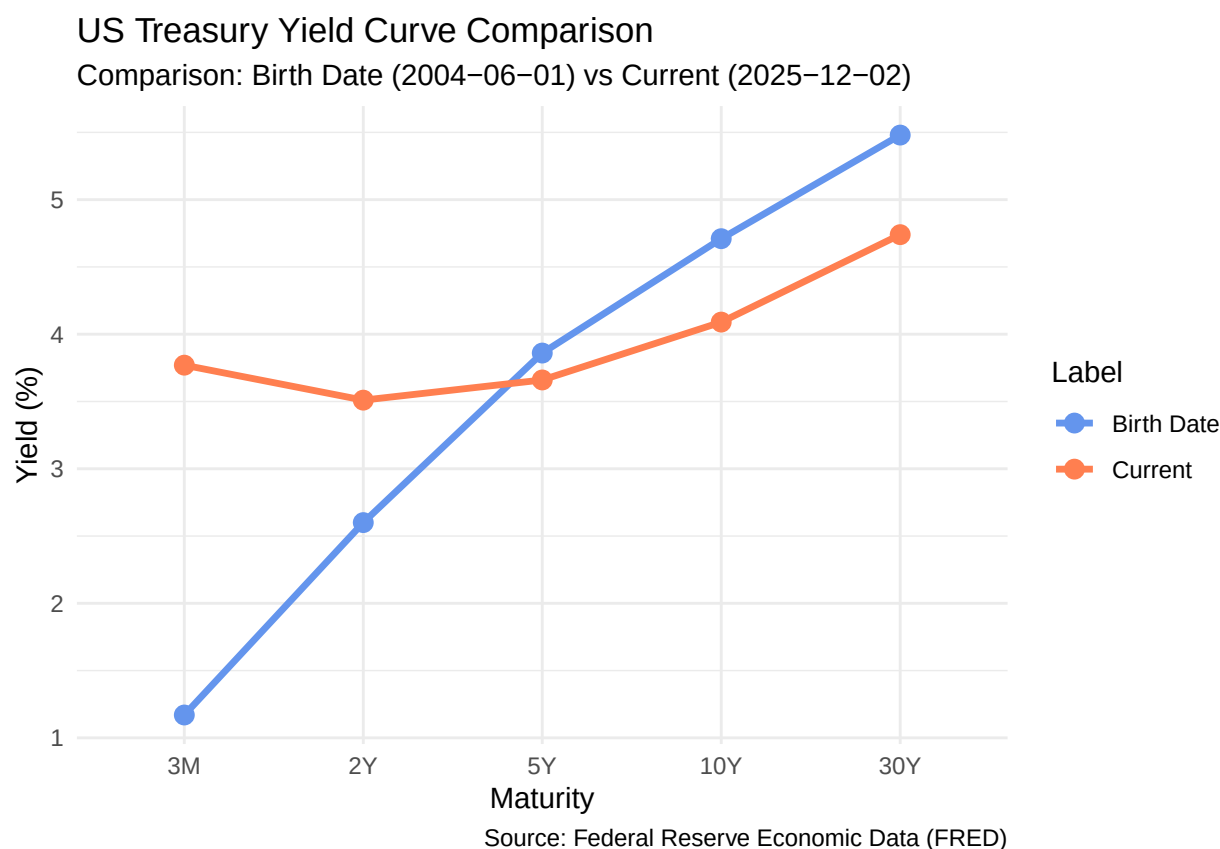
plot_data_long <- plot_data %>%
  pivot_longer(
    cols = c("3M", "2Y", "5Y", "10Y", "30Y"),
    names_to = "Maturity",
    values_to = "Yield"
  ) %>%
  mutate(Maturity = factor(Maturity, levels = c("3M", "2Y", "5Y", "10Y", "30Y")))

ggplot(plot_data_long, aes(x = Maturity, y = Yield, color = Label, group = Label)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  theme_minimal() +
  labs(
```

```

title = "US Treasury Yield Curve Comparison",
subtitle = paste("Comparison: Birth Date (", birth_row$Date, ") vs Current (", current_row$Date, ")"),
y = "Yield (%)",
x = "Maturity",
caption = "Source: Federal Reserve Economic Data (FRED)"
) +
scale_color_manual(values = c("Birth Date" = "cornflowerblue", "Current" = "#FF7F50"))

```



In the graph above, an immediate difference between the two curves can be seen; the relatively normal yield curve on my birthday, with the opposite for current day.

With that being said, any individual would have been better off investing in long-term bonds on my birthday (5.48% @ 30Y) compared to today (4.74 @ 30Y). The steady growth from short-term to long-term rates indicates that investors at the time were optimistic for strong economic growth and inflation.

As stated before, the current yield curve shows signs of inversion (for the front end of investments). The 3M (three month) yield being at a higher point than the 5Y (five year) yield indicates that the Fed is attempting to minimize inflation by locking away short term expenditures through higher short-term bond rates, essentially cooling demand.

In other words, the interest rates on these short-term or long-term bonds can be seen as attempts by the Federal Reserve to either punish those holding liquid capital (through high yield long-term bonds, as seen in 2004), or vice versa (high yield short-term bonds, as seen today)

```

#CPIAUCSL = Consumer Price Index (Inflation)
#UNRATE = Unemployment Rate

macro_tickers <- c("CPIAUCSL", "UNRATE")

```

```

macro_data <- do.call(merge, lapply(macro_tickers, function(x) {
  getSymbols(x, src = "FRED", auto.assign = FALSE)
}))

macro_monthly <- to.monthly(macro_data, indexAt = "lastof", OHLC = FALSE)

## Warning in to.period(x, "months", indexAt = indexAt, name = name, ...): missing
## values removed from data

macro_df <- data.frame(Date = index(macro_monthly), coredata(macro_monthly))
colnames(macro_df) <- c("Date", "CPI", "Unemployment")

birth_macro <- macro_df %>%
  filter(format(Date, "%Y-%m") == "2004-05") %>%
  mutate(Period = "Birth Date")

current_macro <- macro_df %>%
  tail(1) %>%
  mutate(Period = "Current")

macro_table_data <- bind_rows(birth_macro, current_macro) %>%
  select(Period, Date, CPI, Unemployment)

table_data <- bind_rows(birth_row, current_row) %>%
  select(Label, Date, `3M`, `2Y`, `5Y`, `10Y`, `30Y`)

table_data %>%
  kbl(caption = "Yield Curve Data Comparison", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position"), position = "center") %>%
  add_header_above(c(" " = 2, "US Treasury Maturities" = 5)) %>%
  row_spec(2, bold = TRUE, color = "#FF7F50")

```

Table 1: Yield Curve Data Comparison

Label	Date	US Treasury Maturities				
		3M	2Y	5Y	10Y	30Y
Birth Date	2004-06-01	1.17	2.60	3.86	4.71	5.48
Current	2025-12-02	3.77	3.51	3.66	4.09	4.74

```

macro_table_data %>%
  kbl(caption = "Macroeconomic Indicators: Then vs. Now", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position"), position = "center") %>%
  add_header_above(c(" " = 2, "Economic Context" = 2)) %>%
  column_spec(4, bold = TRUE)

```

The shape of the yield curves can be further explained by the macroeconomic indicators in Table 2: in 2004, the upward sloping yield curve aligns with the unemployment rate of 5.6%; it makes sense that the government kept short-term yields low (1.17%)- to stimulate investment in job creation AND borrowing, thus driving inflation rates.

Table 2: Macroeconomic Indicators: Then vs. Now

Period	Date	Economic Context	
		CPI	Unemployment
Birth Date	2004-05-31	188.200	5.6
Current	2025-09-30	324.368	4.4

The current yield curve coincides with a lower unemployment rate of 4.4%. While the labor market is stronger than in 2004, the elevated short-term yields (3.77%) indicate that the central bank's priority has shifted from job creation to controlling inflation.